

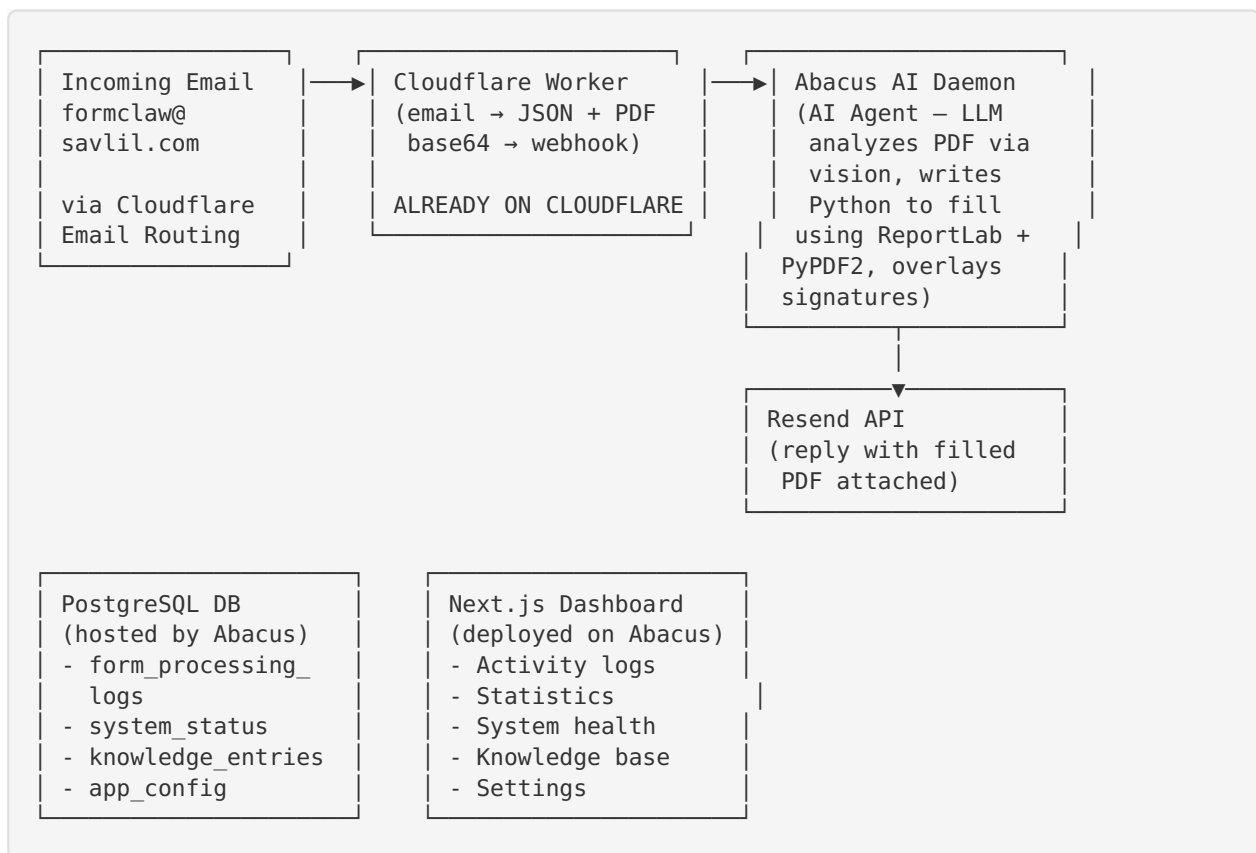
Form Claw — Cloudflare Workers Migration Plan

Date: 2026-07-09

Author: Form Claw Project

Status: DRAFT — awaiting review

1. Current Architecture (Abacus AI + Cloudflare hybrid)



Key insight: The “brain” is an LLM agent

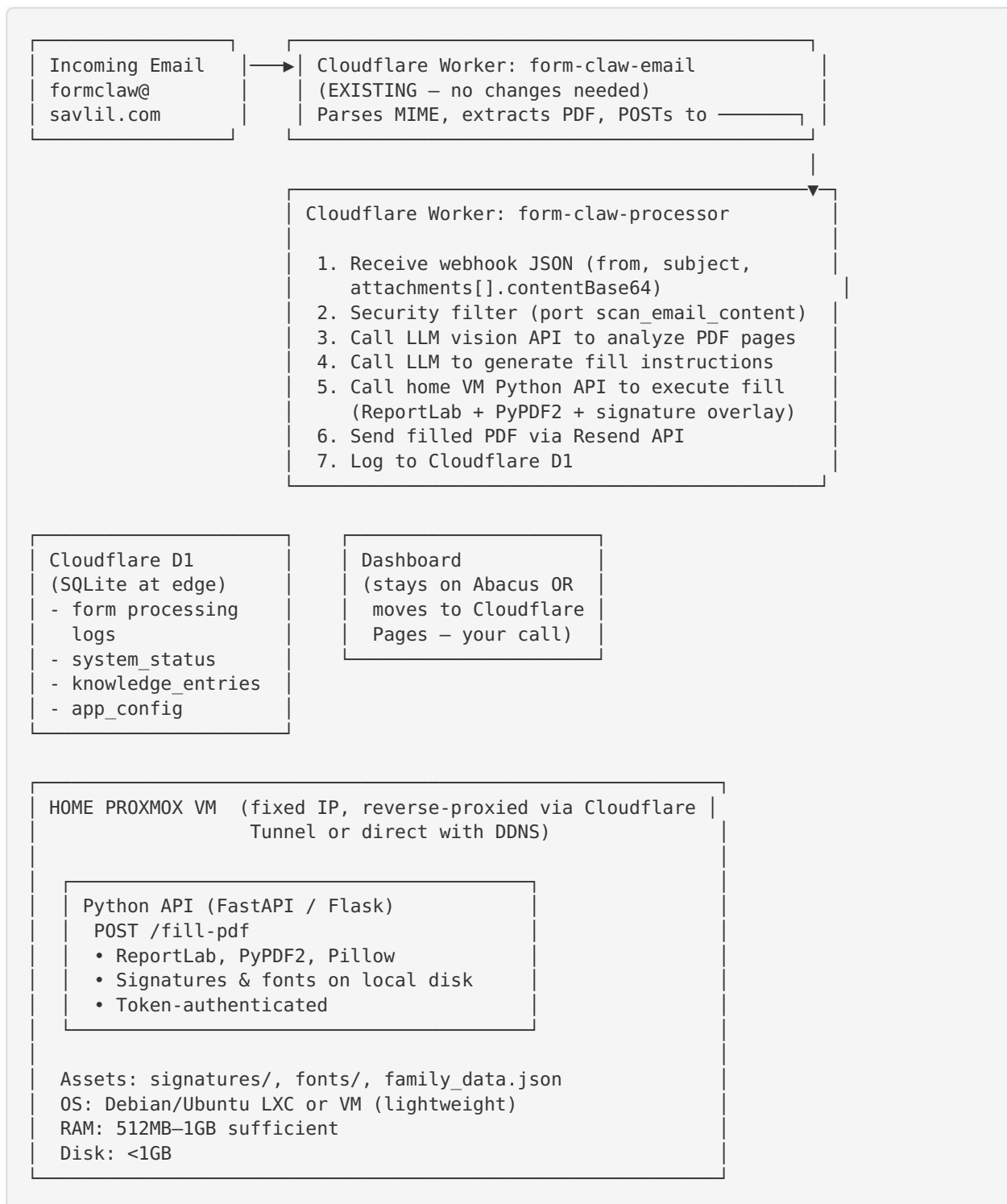
The form processor is NOT a deterministic script. It is an **AI agent** that:

1. Receives a PDF (base64) in the webhook payload
2. Uses **LLM vision** to analyze the PDF form (identify field labels, coordinates, checkboxes)
3. Maps fields to family data from the knowledge base
4. **Writes and executes Python code** at runtime using ReportLab + PyPDF2 to:
 - Place text at precise pixel coordinates
 - Handle Hebrew RTL text
 - Split ID numbers into individual digit boxes
 - Draw ellipses around selected options
 - Overlay transparent signature PNGs
5. Returns the filled PDF

This means the migration is NOT about porting a fixed codebase — it's about:

- Providing an LLM API that supports **vision** (PDF page → image analysis)
- Providing a **code execution sandbox** (Python with ReportLab, PyPDF2, Pillow)
- Providing **storage** for signatures, fonts, family data, and output PDFs

2. Target Architecture (Cloudflare + self-hosted VM)



3. Component-by-Component Migration

3.1 Email Intake Worker (form-claw-email)

Status:  Already on Cloudflare

Changes needed: Point `WEBHOOK_URL` to the new processor worker instead of Abacus daemon

Effort: Config change only (update Cloudflare secret)

3.2 Form Processor Worker (form-claw-processor) — THE HARD PART

This is the core challenge. The current processor is an AI agent with:

- Full Python runtime (ReportLab, PyPDF2, Pillow, tesseract)
- Multi-step LLM reasoning with vision
- File system access (read/write PDFs, PNGs)

Cloudflare Workers limitations:

- No Python runtime (Workers run V8/JavaScript/WASM)
- 30-second CPU time limit (free) / 15-minute wall-clock (paid, with Cron Triggers)
- No file system — must use R2 for storage
- Memory limit: 128MB

Proposed approach: The Worker becomes an **orchestrator** that calls your home VM + LLM API:

```
form-claw-processor (JS/TS Worker)
├── LLM API (vision + reasoning)
│   ├── Analyze PDF pages as images
│   ├── Determine field coordinates & values
│   └── Generate fill instructions JSON
├── Home VM Python API (self-hosted on Proxmox)
│   ├── Receives fill instructions + PDF + signature refs
│   ├── Executes ReportLab/PyPDF2 fill
│   ├── Returns filled PDF base64
│   └── Assets (signatures, fonts) stored locally on the VM
├── R2 (Cloudflare object storage)
│   └── Store filled PDFs temporarily (optional backup)
├── Resend API (send reply – already working)
└── Cloudflare D1 (SQLite)
    ├── Log processing events
    ├── System status
    └── Knowledge base
```

3.3 Database — Cloudflare D1

Current: Abacus-hosted PostgreSQL

Target: [Cloudflare D1](https://developers.cloudflare.com/d1/) (<https://developers.cloudflare.com/d1/>) (SQLite at the edge)

Why D1: Native Cloudflare integration — zero latency from Workers, no external connection strings, free tier generous (5GB storage, 5M reads/day, 100K writes/day), zero cold-start

Migration work:

- Rewrite Prisma schema → D1 SQL schema (SQLite dialect: no `@db.Text`, no enums, `REAL` instead of `Decimal`, `TEXT` for DateTime stored as ISO strings)

- Replace Prisma ORM calls in the Worker with D1 prepared statements (direct `env.DB.prepare()`)
- Dashboard can either:
- (a) Keep Prisma and read from D1 via the Cloudflare REST API (`/client/v4/accounts/.../d1/data-base/.../query`), or
- (b) Stay on Abacus Postgres and sync data via a lightweight D1→Postgres replication (a Cron Trigger that pushes new rows)

D1 Schema (SQLite equivalent of current Prisma models):

```

CREATE TABLE form_processing_log (
  id TEXT PRIMARY KEY DEFAULT (lower(hex(randblob(16)))),
  sender_email TEXT,
  sender_name TEXT,
  subject TEXT,
  received_at TEXT DEFAULT (datetime('now')),
  processing_status TEXT DEFAULT 'pending',
  processing_time_seconds REAL,
  target_person TEXT,
  error_type TEXT,
  error_message TEXT,
  filled_pdf_url TEXT,
  original_pdf_url TEXT,
  ai_notes TEXT
);

CREATE TABLE system_status (
  id INTEGER PRIMARY KEY DEFAULT 1,
  email_source TEXT DEFAULT 'cloudflare',
  webhook_enabled INTEGER DEFAULT 1,
  form_process_status TEXT DEFAULT 'ok',
  last_form_processed TEXT,
  last_error_at TEXT,
  last_cloudflare_email TEXT,
  updated_at TEXT DEFAULT (datetime('now'))
);

CREATE TABLE knowledge_entry (
  id TEXT PRIMARY KEY DEFAULT (lower(hex(randblob(16)))),
  key TEXT NOT NULL,
  value TEXT NOT NULL,
  category TEXT DEFAULT 'general',
  language TEXT DEFAULT 'he',
  applies_to_person TEXT,
  source TEXT DEFAULT 'manual',
  is_active INTEGER DEFAULT 1,
  created_at TEXT DEFAULT (datetime('now')),
  updated_at TEXT DEFAULT (datetime('now'))
);

CREATE TABLE app_config (
  id TEXT PRIMARY KEY DEFAULT (lower(hex(randblob(16)))),
  key TEXT UNIQUE NOT NULL,
  value TEXT NOT NULL,
  label TEXT,
  category TEXT DEFAULT 'general',
  updated_at TEXT DEFAULT (datetime('now'))
);

CREATE INDEX idx_log_received ON form_processing_log(received_at);
CREATE INDEX idx_log_status ON form_processing_log(processing_status);
CREATE INDEX idx_knowledge_active ON knowledge_entry(is_active, category);

```

3.4 Dashboard

Option A (recommended for phase 1): Keep on Abacus

- Dashboard continues to work as-is, using Abacus Postgres
- Add a D1 sync: Processor Worker writes to D1, a Cron Trigger pushes new rows to Abacus Postgres hourly
- Zero dashboard code changes

Option B (full migration): Move to Cloudflare Pages + D1

- Next.js on Pages has limitations (edge runtime, no native Prisma)
- Replace Prisma with `better-sqlite3` client via D1 HTTP API or Drizzle ORM with D1 driver
- Auth: Cloudflare Access (free for ≤ 50 users) replaces NextAuth + Google SSO
- Significant rewrite but zero external dependencies

Option C (self-host dashboard on Proxmox VM too)

- Run the Next.js dashboard directly on the home VM
- Use D1 HTTP API for reads or run a local SQLite replica synced from D1
- Expose via Cloudflare Tunnel (dashboard.savlil.com $\rightarrow$ VM:3000)
- Full control, but adds VM maintenance burden

Recommendation: Start with **Option A** (keep on Abacus), migrate to **Option C** later if desired.

3.5 Health Monitoring

- The `/api/health/check` route stays with the dashboard (wherever it lives)
- The Warning Investigator can run as a Cloudflare Cron Trigger
- Alerts continue via Resend API

4. LLM API Options

The form processor needs:

1. **Vision capability:** Send PDF pages as images, get field analysis
2. **Reasoning:** Multi-step thinking to map family data to detected fields
3. **Structured output:** Return JSON with field coordinates and values
4. **Hebrew language support:** Understand Hebrew form labels

Option A: Abacus AI ChatLLM API (current)

Endpoint: `https://apps.abacus.ai/api/v0/chat/completions`

Key: `ABACUSAI_API_KEY` (already configured)

Models: Routes to best available (GPT-4o, Claude, Gemini, etc.)

Vision: ☒ Supported via `modalities: ["image"]` or base64 image in messages

Hebrew: ☒ Excellent

Cost: Included in Abacus subscription (you're already paying)

Pros:

- Already configured and working
- No additional API key needed
- Model routing optimizes cost/quality

Cons:

- Dependency on Abacus AI platform
- Can't use if you leave Abacus entirely

Integration from Cloudflare Worker:

```
const response = await fetch('https://apps.abacus.ai/api/v0/chat/completions', {
  method: 'POST',
  headers: {
    'Authorization': `Bearer ${env.ABACUSAI_API_KEY}`,
    'Content-Type': 'application/json'
  },
  body: JSON.stringify({
    messages: [
      {
        role: 'user',
        content: [
          { type: 'text', text: 'Analyze this PDF form...' },
          { type: 'image_url', image_url: { url: `data:image/png;base64,${pageImage-
Base64}` } }
        ]
      }
    ],
    model: 'gpt-4o' // or let RouteLLM choose
  })
});
```

Option B: xAI Grok API

Endpoint: `https://api.x.ai/v1/chat/completions`

Key: Get from console.x.ai (<https://console.x.ai>)

Models: `grok-2-vision-1212` (vision), `grok-2-1212` (text)

Vision: ☒ Supported (OpenAI-compatible format)

Hebrew: ☒ Good (not as extensively tested as GPT-4o for Hebrew)

Pricing (as of 2026):

| Model | Input | Output |

|-----|-----|-----|

| grok-2-vision | \$2.00 / 1M tokens | \$10.00 / 1M tokens |

| grok-2 | \$2.00 / 1M tokens | \$10.00 / 1M tokens |

Integration from Cloudflare Worker:

```
const response = await fetch('https://api.x.ai/v1/chat/completions', {
  method: 'POST',
  headers: {
    'Authorization': `Bearer ${env.XAI_API_KEY}`,
    'Content-Type': 'application/json'
  },
  body: JSON.stringify({
    model: 'grok-2-vision-1212',
    messages: [
      {
        role: 'user',
        content: [
          { type: 'text', text: 'Analyze this PDF form...' },
          { type: 'image_url', image_url: { url: `data:image/png;base64,${pageImage-
Base64}` } }
        ]
      }
    ]
  })
});
```

Pros:

- OpenAI-compatible API (easy swap)
- Competitive pricing
- No platform lock-in

Cons:

- Requires separate API key & billing
- Hebrew OCR quality may vary vs GPT-4o
- Grok vision is newer, less battle-tested for precise coordinate extraction

5. The Python Sandbox — Self-Hosted on Proxmox

The form filler needs Python with:

- `reportlab` — PDF overlay creation with precise coordinates
- `PyPDF2` — PDF merging
- `Pillow` — Signature image processing (RGBA transparency)
- Hebrew text shaping (RTL)

Cloudflare Workers cannot run Python natively. But you have a home server — so we self-host.

5a. Home VM Python API (recommended)

Run a lightweight Python API on a **Proxmox LXC container or VM** at home.

VM / LXC Requirements

Requirement	Minimum	Recommended
OS	Debian 12 / Ubuntu 22.04 (LXC or VM)	Ubuntu 24.04 LXC (lowest overhead)
CPU	1 vCPU	2 vCPU (PDF rendering benefits from extra core)
RAM	512 MB	1 GB (ReportLab + Pillow peak during render)
Disk	2 GB	5 GB (OS + Python + deps + fonts + signatures + logs)
Network	Fixed IP, port forwarded (443 → VM)	Cloudflare Tunnel (zero port forwarding)
Python	3.10+	3.12

Setup Steps

```
# 1. Create LXC container in Proxmox (or use an existing VM)
#   Proxmox UI → Create CT → Ubuntu 24.04, 1GB RAM, 5GB disk

# 2. Inside the container:
apt update && apt install -y python3 python3-pip python3-venv

# 3. Create project directory
mkdir -p /opt/form-claw-pdf
cd /opt/form-claw-pdf
python3 -m venv venv
source venv/bin/activate

# 4. Install dependencies
pip install fastapi uvicorn reportlab PyPDF2 Pillow python-multipart

# 5. Copy assets
mkdir -p assets/signatures assets/fonts
# scp or git clone from zeevrussak/form-claw → services/pdf-filler/
# Copy signature PNGs and font TTFs into assets/

# 6. Create systemd service
cat > /etc/systemd/system/form-claw-pdf.service << 'EOF'
[Unit]
Description=Form Claw PDF Filler API
After=network.target

[Service]
Type=simple
User=root
WorkingDirectory=/opt/form-claw-pdf
ExecStart=/opt/form-claw-pdf/venv/bin/uvicorn app:app --host 0.0.0.0 --port 8787
Restart=always
RestartSec=5
Environment=PDF_SERVICE_TOKEN=<your-secret-token>

[Install]
WantedBy=multi-user.target
EOF

systemctl enable --now form-claw-pdf
```

FastAPI Application Skeleton

```
# /opt/form-claw-pdf/app.py
from fastapi import FastAPI, Header, HTTPException
from pydantic import BaseModel
import base64, os

app = FastAPI(title="Form Claw PDF Filler")
TOKEN = os.environ.get("PDF_SERVICE_TOKEN", "change-me")

class FillRequest(BaseModel):
    original_pdf_b64: str
    fill_instructions: list[dict] # [{x, y, text, font, size, color}, ...]
    signature_id: str | None = None # "zeev" or "keren"
    signer_name: str | None = None
    date_str: str | None = None

class FillResponse(BaseModel):
    filled_pdf_b64: str
    pages_processed: int

@app.post("/fill-pdf", response_model=FillResponse)
async def fill_pdf(req: FillRequest, authorization: str = Header(...)):
    if authorization != f"Bearer {TOKEN}":
        raise HTTPException(401, "Unauthorized")
    # ... ReportLab + PyPDF2 fill logic here (ported from fill_fnx_form.py) ...
    return FillResponse(filled_pdf_b64="...", pages_processed=1)

@app.get("/health")
async def health():
    return {"status": "ok"}
```

Exposing the VM to the Internet

Option 1: Cloudflare Tunnel (recommended — zero port forwarding)

```
# Install cloudflared on the VM
curl -L https://github.com/cloudflare/cloudflared/releases/latest/download/
cloudflared-linux-amd64.deb -o cloudflared.deb
dpkg -i cloudflared.deb

# Login and create tunnel
cloudflared tunnel login
cloudflared tunnel create form-claw-pdf

# Configure
cat > /etc/cloudflared/config.yml << EOF
tunnel: <TUNNEL_ID>
credentials-file: /root/.cloudflared/<TUNNEL_ID>.json
ingress:
  - hostname: pdf.savlil.com
    service: http://localhost:8787
  - service: http_status:404
EOF

# DNS: add CNAME pdf.savlil.com → <TUNNEL_ID>.cfargotunnel.com
cloudflared tunnel route dns form-claw-pdf pdf.savlil.com

# Run as service
cloudflared service install
systemctl enable --now cloudflared
```

Result: `https://pdf.savlil.com/fill-pdf` is publicly reachable, TLS terminated by Cloudflare, no ports exposed.

Option 2: Direct with fixed IP

- Port-forward 443 → VM port 8787 on your router
- Use Caddy or Nginx as reverse proxy with Let's Encrypt TLS
- DNS A record: `pdf.savlil.com → <your-fixed-IP>`
- Less secure (IP exposed) but works without Cloudflare Tunnel

Recommendation: Use Cloudflare Tunnel — it's free, secure, and integrates with your existing Cloudflare setup.

Auto-Update from GitHub

```
# /opt/form-claw-pdf/update.sh
#!/bin/bash
cd /opt/form-claw-pdf
git pull origin main
source venv/bin/activate
pip install -r requirements.txt --quiet
systemctl restart form-claw-pdf
echo "Updated at $(date)"
```

Add a cron job or GitHub Actions webhook to trigger updates:

```
# Check for updates every hour
0 * * * * /opt/form-claw-pdf/update.sh >> /var/log/form-claw-update.log 2>&1
```

Or use a GitHub Actions deploy step that SSHs into the VM (see Section 7).

5b. pdf-lib (JavaScript, runs in Worker) — NOT RECOMMENDED

Use [pdf-lib](https://pdf-lib.js.org/) (https://pdf-lib.js.org/) for basic PDF manipulation inside the Worker itself.

Limitations:

- ❌ No Hebrew RTL text shaping
- ❌ No transparent PNG overlay with the same quality as ReportLab
- ❌ No ellipse drawing for OR/slash selection

Verdict: Not suitable for the current form complexity.

5c. Cloudflare Workers + Python (experimental) — NOT READY

Cloudflare Python Workers (beta) don't support ReportLab/PyPDF2. Not viable.

6. Storage: Split Strategy

Static assets → Home VM local disk (simplest)

Since the Python API runs on your Proxmox VM, static assets live there directly — no need to fetch from R2 on every request:

Asset	VM Path
Ze'ev's signature	/opt/form-claw-pdf/assets/signatures/zeev.png
Keren's signature	/opt/form-claw-pdf/assets/signatures/keren.png
Hebrew font	/opt/form-claw-pdf/assets/fonts/Ft-PilKahol2.ttf
English font	/opt/form-claw-pdf/assets/fonts/Playzone.ttf
Family data	/opt/form-claw-pdf/assets/family_data.json

Output PDFs → Cloudflare R2 (optional backup)

Asset	R2 Key
Filled PDFs	output/<timestamp>/<filename>_filled.pdf

R2 is useful if you want the dashboard to serve download links for filled PDFs. Otherwise, filled PDFs are attached directly to the Resend reply email and don't need persistent storage.

R2 Cost: Free tier — 10GB storage, 10M reads/month, 1M writes/month. More than enough.

7. GitHub Integration & CI/CD

All worker code will be sourced from the **existing GitHub repo** ([zeevrussak/form-claw](#)).

Repository Structure (proposed additions)

```

form-claw/
├── workers/
│   ├── email-intake/                # existing Cloudflare Email Worker
│   │   ├── src/index.ts
│   │   ├── wrangler.toml
│   │   └── package.json
│   ├── form-processor/              # NEW: orchestrator worker
│   │   ├── src/
│   │   │   ├── index.ts             # main handler
│   │   │   ├── llm.ts               # LLM API client (ChatLLM or Grok)
│   │   │   ├── security.ts          # port of security_filter.py
│   │   │   ├── pdf-service.ts       # calls home VM Python API
│   │   │   ├── db.ts                # D1 database client
│   │   ├── wrangler.toml
│   │   └── package.json
│   └── analytics/                   # OPTIONAL Phase 6: GraphQL analytics
│       ├── src/index.ts
│       ├── wrangler.toml
│       └── package.json
├── services/
│   └── pdf-filler/                  # Python API (runs on home Proxmox VM)
│       ├── app.py
│       ├── requirements.txt
│       └── assets/
│           ├── signatures/
│           ├── fonts/
│           └── family_data.json
├── d1/
│   └── schema.sql                  # D1 schema (source of truth)
└── nextjs_space/                   # existing dashboard (stays on Abacus)

```

Cloudflare Wrangler Configuration

```

# workers/form-processor/wrangler.toml
name = "form-claw-processor"
main = "src/index.ts"
compatibility_date = "2026-01-01"

[[d1_databases]]
binding = "DB"
database_name = "form-claw"
database_id = "<your-d1-database-id>"

[[r2_buckets]]
binding = "ASSETS"
bucket_name = "form-claw-assets"

[vars]
PDF_SERVICE_URL = "https://pdf.savlil.com"
DASHBOARD_URL = "https://form-claw.abacusai.app"
LLM_PROVIDER = "chatllm"
WHITELISTED_SENDERS = "k6622024@gmail.com,2396119@gmail.com"

```

Cloudflare Secrets (via `wrangler secret put`)

Secret Name	Purpose	Source
<code>RESEND_API_KEY</code>	Send reply emails	Resend dashboard
<code>LLM_API_KEY</code>	LLM API authentication	See Option A or B below
<code>PDF_SERVICE_TOKEN</code>	Auth for home VM Python API	Self-generated (e.g., <code>openssl rand -hex 32</code>)
<code>HEARTBEAT_TOKEN</code>	Dashboard health reporting	Existing value

Cloudflare Variables (non-secret, in `wrangler.toml` `[vars]`)

Variable	Value
<code>PDF_SERVICE_URL</code>	<code>https://pdf.savlil.com</code>
<code>DASHBOARD_URL</code>	<code>https://form-claw.abacusai.app</code>
<code>LLM_PROVIDER</code>	<code>chatllm</code> or <code>grok</code>
<code>WHITELISTED_SENDERS</code>	<code>k6622024@gmail.com,2396119@gmail.com,...</code>

CI/CD via GitHub Actions

```
# .github/workflows/deploy-workers.yml
name: Deploy Cloudflare Workers
on:
  push:
    branches: [main]
    paths: ['workers/**']
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: cloudflare/wrangler-action@v3
        with:
          apiToken: ${ secrets.CLOUDFLARE_API_TOKEN }
          workingDirectory: workers/form-processor

# .github/workflows/deploy-pdf-service.yml
name: Deploy PDF Service to Home VM
on:
  push:
    branches: [main]
    paths: ['services/pdf-filler/**']
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Deploy via SSH
        uses: appleboy/ssh-action@v1
        with:
          host: pdf.savlil.com # or your fixed IP
          username: root
          key: ${ secrets.VM_SSH_PRIVATE_KEY }
          script: |
            cd /opt/form-claw-pdf
            git pull origin main
            source venv/bin/activate
            pip install -r requirements.txt --quiet
            systemctl restart form-claw-pdf
            echo "Deployed at $(date)"
```

GitHub Secrets Required

Secret	Purpose
CLOUDFLARE_API_TOKEN	Deploy Workers via Wrangler
VM_SSH_PRIVATE_KEY	SSH deploy to Proxmox VM

8. Cost Estimate — 2 forms/day

Per-form processing breakdown

Each form fill involves approximately:

- **1 LLM vision call** (analyze 1-2 PDF pages as images): ~2,000 input tokens + ~1,500 output tokens per page
- **1 LLM text call** (generate fill instructions): ~3,000 input tokens + ~2,000 output tokens
- **1 Python API call** (fill PDF): CPU time ~3 seconds
- **1 Resend email** (reply with PDF)
- **1 DB write** (log entry)

Estimated tokens per form: ~5,000 input + ~3,500 output (single page form)

Monthly cost at 2 forms/day (60 forms/month)

Component	Option A: ChatLLM	Option B: Grok (xAI)
LLM API	Included in Abacus sub	~\$0.60 input + ~\$2.10 output = ~\$2.70/mo
Cloudflare Workers	Free tier (100K req/day)	Free tier
Cloudflare D1	Free tier (5GB, 5M reads/day)	Free tier
Cloudflare R2	Free tier (10GB)	Free tier
Cloudflare Email Routing	Free	Free
Resend API	Free tier (100 emails/day)	Free tier
Home VM (Proxmox)	\$0 (your hardware + electricity)	\$0
GitHub	Free	Free
Total monthly	~\$0/mo (Abacus sub covers LLM)	~\$3/mo

Token math for Grok

Per form:

Vision: $2,000 \text{ input} \times \$2/1\text{M} = \$0.004$ + $1,500 \text{ output} \times \$10/1\text{M} = \$0.015$

Text: $3,000 \text{ input} \times \$2/1\text{M} = \$0.006$ + $2,000 \text{ output} \times \$10/1\text{M} = \$0.020$

Total per form: ~\$0.045

Monthly (60 forms): $60 \times \$0.045 = \2.70

Note: Multi-page forms or complex forms requiring multiple LLM rounds could double costs. Estimate assumes typical 1-page Israeli government/HMO forms.

9. Implementation Phases

Phase 1: Foundation (1-2 days)

- [] Create Cloudflare D1 database with schema (see Section 3.3)
- [] Create R2 bucket (for optional output PDF backup)
- [] Set up GitHub repo structure (`workers/` , `services/`)
- [] Set up GitHub Actions for Cloudflare + VM deployment

Phase 2: Home VM Setup (1-2 days)

- [] Provision LXC container on Proxmox (Ubuntu 24.04, 1GB RAM, 5GB disk)
- [] Install Python 3.12, create venv, install deps (FastAPI, ReportLab, PyPDF2, Pillow)
- [] Port `fill_fnx_form.py` + skill logic into stateless FastAPI `/fill-pdf` endpoint
- [] Copy assets (signatures, fonts, family_data.json) into `/opt/form-claw-pdf/assets/`
- [] Set up Cloudflare Tunnel (`pdf.savlil.com` → VM:8787)
- [] Set up systemd service for auto-start
- [] Test with known form + fill instructions via curl

Phase 3: Processor Worker (2-3 days)

- [] Write `form-claw-processor` Worker:
- Webhook handler
- Security filter (port `security_filter.py` to TypeScript)
- LLM client (with provider switch: ChatLLM / Grok)
- PDF service client (calls `https://pdf.savlil.com/fill-pdf`)
- Resend client (port `send_resend_reply.py` logic)
- D1 logging (`env.DB.prepare()` statements)
- Heartbeat reporting
- [] Deploy and wire up email intake Worker

Phase 4: Dashboard Reconnection (1 day)

- [] Add D1→Abacus Postgres sync (Cron Trigger or D1 REST API adapter)
- [] Update health check endpoint for new architecture
- [] Test all dashboard pages with synced data

Phase 5: Cutover & Monitoring (1 day)

- [] Update email intake Worker's `WEBHOOK_URL` to processor Worker
- [] Run end-to-end test (email → fill → reply)
- [] Monitor first 24h of production
- [] Disable old Abacus daemon

Total estimated effort: 6-9 days

10. Risks & Mitigations

Risk	Impact	Mitigation
Worker 30s CPU limit exceeded on complex forms	Processing fails	Use Cloudflare Workers Unbound (no CPU limit, \$0.02/M requests) or offload heavy work to Python service
Hebrew RTL quality degrades with Grok vs GPT-4o	Incorrect field fills	Test both providers with 5+ real forms before switching; keep ChatLLM as fallback
Home VM downtime (power outage, Proxmox reboot)	Processing queued/fails	UPS for Proxmox host; Worker retries with exponential back-off; email stays in Cloudflare queue
Home internet outage	VM unreachable	Cloudflare Tunnel auto-reconnects; forms queue in Worker until VM is back; set 3-retry with 30s delay
PDF page-to-image conversion needed for vision	Workers can't render PDFs	Do PDF→PNG conversion in the home VM Python service; return images to Worker for LLM call
D1 schema drift from Prisma schema	Dashboard data mismatch	Maintain single source-of-truth SQL migration files; apply to both D1 and Prisma schema in same PR

11. Decision Matrix

Question	Recommended Choice
LLM Provider	Start with ChatLLM (free, tested). Switch to Grok if leaving Abacus.
PDF Processing	Self-hosted Python API on Proxmox VM via Cloudflare Tunnel
Database	Cloudflare D1 (SQLite, free tier, native to Workers)
Storage (assets)	Local disk on home VM (signatures, fonts)
Storage (output)	Cloudflare R2 (optional backup for filled PDFs)
Dashboard	Keep on Abacus (phase 1). Self-host on VM later (optional).
CI/CD	GitHub Actions → Wrangler (Workers) + SSH deploy (VM)
Analytics API	Cloudflare D1 + GraphQL (see Section 14)

12. Analytics via GraphQL (optional enhancement)

The current dashboard reads stats via REST endpoints (`/api/stats` , `/api/stats/range`). With D1 as the data source, you could expose a **GraphQL API** on a Cloudflare Worker for analytics — this is a good fit because:

Why GraphQL works well here

- **Flexible queries:** The dashboard already supports date ranges, sender filters, target-person filters, and status filters. GraphQL lets the frontend request exactly the aggregations it needs in a single request instead of multiple REST calls.
- **D1 is SQLite:** GraphQL resolvers map cleanly to SQL queries. No ORM needed — just `env.DB.prepare()` with parameterized queries.
- **Edge performance:** A Worker + D1 GraphQL endpoint runs at the edge with ~0ms DB latency.
- **Schema-first:** GraphQL's typed schema documents the analytics API automatically.

Example GraphQL Schema

```

type Query {
  stats(startDate: String, endDate: String)! StatsOverview!
  dailyStats(startDate: String!, endDate: String!): [DailyStat!]!
  logs(page: Int, limit: Int, status: String, sender: String, search: String)!
  LogPage!
  senderBreakdown(startDate: String, endDate: String)! [SenderStat!]!
  targetBreakdown(startDate: String, endDate: String)! [TargetStat!]!
}

type StatsOverview {
  totalForms: Int!
  successCount: Int!
  failureCount: Int!
  successRate: Float!
  avgProcessingTime: Float
  todayCount: Int!
}

type DailyStat {
  date: String!
  total: Int!
  success: Int!
  failure: Int!
}

type LogPage {
  logs: [FormLog!]!
  total: Int!
  page: Int!
  totalPages: Int!
}

type FormLog {
  id: String!
  senderEmail: String
  subject: String
  receivedAt: String!
  processingStatus: String!
  processingTime: Float
  targetPerson: String
  errorType: String
  errorMessage: String
}

type SenderStat { sender: String!, count: Int! }
type TargetStat { target: String!, count: Int! }

```

Implementation

Use [graphql-yoga](https://the-guild.dev/graphql/yoga-server) (<https://the-guild.dev/graphql/yoga-server>) — it runs natively in Cloudflare Workers:

```
// workers/analytics/src/index.ts
import { createYoga, createSchema } from 'graphql-yoga';

export default {
  fetch(request: Request, env: Env) {
    const yoga = createYoga({
      schema: createSchema({
        typeDefs,
        resolvers: {
          Query: {
            stats: async (_, { startDate, endDate }) => {
              const result = await env.DB.prepare(
                `SELECT COUNT(*) as total,
                  SUM(CASE WHEN processing_status='success' THEN 1 ELSE 0 END) as suc-
cess,
                  SUM(CASE WHEN processing_status='failed' THEN 1 ELSE 0 END) as fail-
ure,
                  AVG(processing_time_seconds) as avg_time
FROM form_processing_log
WHERE received_at BETWEEN ?1 AND ?2`
              ).bind(startDate || '2000-01-01', endDate || '2099-12-31').first();
              return { ...result, successRate: result.total ? result.success / res-
ult.total : 0 };
            },
            // ... other resolvers
          }
        }
      });
    });
    return yoga.fetch(request, env);
  }
};
```

When to add GraphQL

Not in Phase 1. Get the core form-filling pipeline working first. GraphQL is a Phase 6 enhancement — add it once:

1. D1 is populated with real data
2. The dashboard needs more flexible analytics (or you want a public/mobile analytics view)
3. You want to query analytics from multiple clients (dashboard, mobile, CLI)

For Phase 1-5, the existing REST endpoints on the Abacus dashboard are sufficient.

13. What Stays the Same

- ☒ Email intake Cloudflare Worker (already deployed)
- ☒ Resend for outbound email (already configured)
- ☒ GitHub repository as source of truth
- ☒ Sender whitelist logic
- ☒ Security filter logic (ported to TypeScript)
- ☒ Family data structure (family_data.json)
- ☒ Signature assets (PNGs with transparency)
- ☒ Form-filling skill logic (coordinate placement, ellipses, split-digit fields)

14. What Changes

-  Form processor: Abacus AI daemon → Cloudflare Worker + home VM Python API
-  Database: Abacus-hosted PostgreSQL → Cloudflare D1 (SQLite)
-  Asset storage: Abacus VM filesystem → Home VM local disk
-  LLM calls: Abacus built-in agent → Direct API calls (ChatLLM or Grok)
-  Code execution: Abacus agent Python sandbox → Self-hosted FastAPI on Proxmox
-  Analytics (future): REST → GraphQL on Cloudflare Worker